

# Combined Step-Size Subspace Tracking Algorithm Based on Maximum Correntropy Criterion

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**Abstract**—Subspace tracking plays a fundamental role in array signal processing. Among existing algorithms, the gradient descent subspace tracking (GDST) algorithm is attractive due to its low computational complexity. However, it suffers from two limitations: 1) it is very sensitive to impulsive noises and 2) the constant step-size leads to a performance trade-off between convergence speed and steady-state accuracy. To solve the first issue, we incorporate the maximum correntropy criterion (MCC) into GDST and derive the MCC-GDST algorithm with robustness against impulsive noises. Then, for the second limitation, we design a combined step-size (CSS) scheme and develop the CSS-MCC-GDST algorithm. Simulations in scenarios of both subspace estimation and direction-of-arrival (DOA) estimation verify the superiority of the proposed algorithms over their counterparts.

**Index Terms**—Combined step-size, Direction-of-arrival estimation, Gradient descent subspace tracking, Impulsive noise, Maximum correntropy criterion.

## I. INTRODUCTION

As one of the key aspects of modern array signal processing, direction-of-arrival (DOA) estimation has been extensively studied and applied in radar and acoustic systems, among other fields [1]. Representative DOA estimation algorithms, such as multiple signal classification (MUSIC) [2] and the estimation of signal parameters via rotational invariance techniques (ESPRIT) [3], were studied in depth due to their high-resolution capability. However, these algorithms rely on an accurate estimate of the invariant signal subspace. In a dynamic environment, the covariance matrix of the array output is time-varying, which makes it difficult to obtain an accurate covariance estimate from a finite number of snapshots. In response to this challenge, subspace tracking techniques were developed not only to avoid the eigenvalue decomposition (EVD) required by MUSIC and ESPRIT but also to enable real-time update of the signal subspace from streaming data, serving as a foundation for adaptive processing of DOA.

Subspace tracking commonly adopts the idea of formulating the signal subspace extraction as an unconstrained optimization problem. In [4], based on this formulation, two mean square error (MSE)-based algorithms were presented. One is the projection approximation subspace tracking (PAST) algorithm that minimizes a weighted MSE criterion based on recursive least squares (RLS)-style, and the other is the gradient descent subspace tracking (GDST) algorithm for

simplicity of implementation. However, the performance of these MSE-based subspace tracking algorithms would degrade severely in the presence of impulsive noises [5]. In practice, such impulsive interferences often arise from intermittent electromagnetic interference, switching transients, and sensor outliers [6] [7], which are commonly modeled by a Gaussian-mixture process [8] or a symmetric  $\alpha$ -stable (S $\alpha$ S) process [9].

To improve subspace tracking under impulsive noises, various robust strategies have been developed. For example, the M-estimator was incorporated into the PAST framework in [10]. By exploiting the outlier-suppression property of correntropy, the maximum correntropy criterion (MCC)-PAST algorithm was developed by integrating the MCC into PAST, with an automatically adjusted forgetting factor (FF) to enhance tracking ability under fast subspace variations [11]. In addition, the robust subspace tracking adaptive projection subgradient (RSTAPS) algorithm adopted set-theoretic projections onto data-consistent convex sets and achieved good robustness by detecting outliers while predicting missing entries [12]. Nevertheless, two limitations remain: relatively high complexity due to RLS-type recursions or repeated inner iterations, and sensitivity to the FF setting. Motivated by this, we focus on the GDST framework with lower complexity. However, GDST has not been well studied under impulsive noises and its constant step-size leads to a convergence–accuracy trade-off. Accordingly, our contributions are summarized as follows.

1) To suppress the negative effect of impulsive noises, the MCC is integrated into the GDST framework, yielding the MCC-GDST algorithm applicable for the DOA estimation.

2) We design a combined step-size (CSS) scheme and further incorporate a projection-energy-ratio-based detection together with a reset mechanism, based on which the proposed CSS-MCC-GDST algorithm achieves a more favorable balance between convergence speed and steady-state accuracy while improving the re-tracking capability under abrupt subspace changes.

**Notations:** The superscript  $(\cdot)^H$  denotes the Hermitian transpose. The operator  $\Re(\cdot)$  extracts the real part of a complex-valued quantity, whereas  $|\cdot|$  indicates the absolute value. The notation  $\|\cdot\|_2$  represents the  $\ell_2$ -norm, and  $\mathbb{E}[\cdot]$  stands for the expectation of random variables. The symbol  $\mathbb{C}^{N \times r}$  refers to the set of  $N \times r$  complex-valued matrices.

## II. SUBSPACE TRACKING MODEL AND GDST ALGORITHM

We consider  $r$  narrowband signals  $\{s_k(t)\}_{k=1}^r$  impinging on an array with  $N$  sensors. At the  $t$ -th snapshot, the received data vector  $\mathbf{x}(t) \in \mathbb{C}^{N \times 1}$  can be modeled as

$$\mathbf{x}(t) = \mathbf{A}\mathbf{s}(t) + \mathbf{n}(t), \quad (1)$$

where  $\mathbf{A} = [\mathbf{a}(\omega_1), \mathbf{a}(\omega_2), \dots, \mathbf{a}(\omega_r)] \in \mathbb{C}^{N \times r}$  is the steering matrix with  $\mathbf{a}(\omega_k) \in \mathbb{C}^{N \times 1}$  being the steering vector of the  $k$ -th signal,  $\mathbf{s}(t) = [s_1(t), s_2(t), \dots, s_r(t)]^T \in \mathbb{C}^{r \times 1}$  is the narrow-band source vector and  $\mathbf{n}(t) \in \mathbb{C}^{N \times 1}$  denotes the additive noise vector.

Conventionally, as in the MUSIC, the signal subspace is obtained from the EVD of the covariance matrix of  $\mathbf{x}(t)$ , defined as

$$\begin{aligned} \mathbf{R}_{xx} &= \mathbb{E}[\mathbf{x}(t)\mathbf{x}^H(t)] = \mathbf{A}\mathbf{R}_{ss}\mathbf{A}^H + \sigma_n^2 \mathbf{I} \\ &= \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_n \mathbf{\Lambda}_n \mathbf{U}_n^H, \end{aligned} \quad (2)$$

where  $\mathbf{R}_{ss} = \mathbb{E}[\mathbf{s}(t)\mathbf{s}^H(t)] \in \mathbb{C}^{r \times r}$  is the covariance matrix of  $\mathbf{s}(t)$ ,  $\sigma_n^2$  denotes the variance for the white noise;  $\mathbf{U}_s$  is formed by the eigenvectors associated with the  $r$  largest eigenvalues of  $\mathbf{R}_{xx}$  and it spans the signal subspace,  $\mathbf{U}_n$  is formed by the remaining  $N - r$  eigenvectors and it spans the noise subspace;  $\mathbf{\Lambda}_s$  and  $\mathbf{\Lambda}_n$  are diagonal matrices containing the  $r$  largest and the remaining  $N - r$  eigenvalues of  $\mathbf{R}_{xx}$ , respectively.

However, in dynamic environments,  $\mathbf{R}_{xx}$  varies over time, thus performing an EVD based on data of multiple snapshots becomes prohibitively costly. In contrast, subspace tracking algorithms recursively update a basis matrix  $\mathbf{W}(t) \in \mathbb{C}^{N \times r}$  of the signal subspace, using the current observation  $\mathbf{x}(t)$  and the previous basis matrix  $\mathbf{W}(t-1)$ . Specifically, it was shown in [4] that the signal subspace estimation can be formulated as an unconstrained optimization problem by minimizing the following cost function:

$$J(\mathbf{W}) = \mathbb{E} \left[ \left\| \mathbf{x}(t) - \mathbf{W}\mathbf{W}^H \mathbf{x}(t) \right\|_2^2 \right]. \quad (3)$$

In online implementation, the expectation in (3) is commonly approximated using the instantaneous snapshot, which can motivate a stochastic-gradient update and lead to the GDST algorithm [4] that performs the update

$$\mathbf{W}(t) = \mathbf{W}(t-1) + \mu \mathbf{e}(t) \mathbf{y}^H(t), \quad (4)$$

where  $\mu$  denotes the constant step-size,  $\mathbf{y}(t) \triangleq \mathbf{W}^H(t-1)\mathbf{x}(t)$  represents the projection approximation and  $\mathbf{e}(t) \triangleq \mathbf{x}(t) - \mathbf{W}(t-1)\mathbf{y}(t)$  indicates the error vector.

In practice,  $\mathbf{n}(t)$  may deviate from Gaussianity and exhibit impulsive behavior in the time-domain samples, e.g., the S $\alpha$ S noise, whose characteristic function is given by

$$\phi(\omega) = \exp(-\gamma|\omega|^\alpha), \quad (5)$$

where  $0 < \alpha \leq 2$  is the characteristic exponent and  $\gamma$  is the scale parameter. When  $\alpha = 2$ , the additive noise  $\mathbf{n}(t)$  reduces to the Gaussian case, whereas smaller  $\alpha$  indicates

heavier tails and stronger impulsiveness, yielding more frequent large-amplitude outliers. Under such noise conditions, the conventional GDST algorithm would suffer from severe performance degradation.

## III. PROPOSED CSS-MCC-GDST ALGORITHM

In this section, the MCC is incorporated into GDST to develop the MCC-GDST algorithm. Then, a CSS strategy is derived to improve convergence speed and steady-state accuracy.

### A. Maximum Correntropy Criterion

The correntropy quantifies the similarity between two random variables  $X$  and  $Y$  in a kernel-induced measure, which is formulated as

$$V_\sigma(X, Y) = \mathbb{E}[\kappa_\sigma(X - Y)], \quad (6)$$

where  $\kappa_\sigma(X - Y)$  is a kernel function. Typically, the Gaussian kernel is adopted:

$$\kappa_\sigma(X - Y) = \exp\left(-\frac{|X - Y|^2}{2\sigma^2}\right), \quad (7)$$

and  $\sigma$  denotes the kernel width.

**Remark 1:** When  $X$  is close to  $Y$ , the difference  $|X - Y|$  will be small so that  $\kappa_\sigma(X - Y)$  will stay close to one. When impulsive disturbances occur in either  $X$  or  $Y$ , the difference  $|X - Y|$  becomes large, which means that  $\kappa_\sigma(X - Y)$  will rapidly decay toward zero. Building on such properties, the MCC aims to maximize the correntropy, thereby ensuring that samples with small discrepancies dominate the optimization.

### B. Derivation of MCC-GDST

Benefited from the impulsive-noise suppression property of MCC discussed in Remark 1, we construct the MCC-based cost function for subspace tracking:

$$J_{\text{MCC}}(\mathbf{W}) = \mathbb{E} \left[ \kappa_\sigma(0) - \kappa_\sigma \left( \left\| \mathbf{x}(t) - \mathbf{W}\mathbf{W}^H \mathbf{x}(t) \right\|_2 \right) \right]. \quad (8)$$

Applying the gradient descent optimization to minimize (8), it yields the subspace recursion for the MCC-GDST algorithm as

$$\mathbf{W}(t) = \mathbf{W}(t-1) + \mu \kappa_\sigma(\|\mathbf{e}(t)\|_2) \mathbf{e}(t) \mathbf{y}^H(t). \quad (9)$$

Since the update term in (9) is directly affected by the instantaneous projection energy, we adopt a power-normalized step-size strategy, akin to the normalized LMS, by normalizing the update with  $\|\mathbf{y}(t)\|_2^2$ . In this way, the recursion becomes less sensitive to signal-energy fluctuations. Accordingly, (9) is rewritten as

$$\mathbf{W}(t) = \mathbf{W}(t-1) + \mu \kappa_\sigma(\|\mathbf{e}(t)\|_2) \frac{\mathbf{e}(t) \mathbf{y}^H(t)}{\|\mathbf{y}(t)\|_2^2}. \quad (10)$$

**Remark 2:** Under impulsive noises,  $\|\mathbf{e}(t)\|_2$  can become large, driving  $\kappa_\sigma(\|\mathbf{e}(t)\|_2)$  toward zero and thereby attenuating the instantaneous update of the MCC-GDST algorithm. The impulse-suppression strength of the MCC-GDST algorithm depends on the kernel width  $\sigma$ : as  $\sigma \rightarrow 0$ , the MCC-GDST

algorithm barely updates, while as  $\sigma \rightarrow \infty$ , it reduces to the conventional GDST algorithm and loses its robustness. For the appropriate selection of  $\sigma$ , interested readers may refer to [13], [14]. Importantly, the constant step-size  $\mu$  in the MCC-GDST algorithm induces a trade-off between convergence speed and steady-state accuracy.

### C. Combined Step-size Strategy

To address the trade-off issue stated in Remark 2, we design a combined step-size  $\mu(t)$  to replace the constant step-size  $\mu$  and reformulate (10) as

$$\mathbf{W}(t) = \mathbf{W}(t-1) + \mu(t)\kappa_\sigma(\|\mathbf{e}(t)\|_2) \frac{\mathbf{e}(t)\mathbf{y}^H(t)}{\|\mathbf{y}(t)\|_2^2}, \quad (11)$$

where  $\mu(t) = \lambda(t)\mu_1 + (1 - \lambda(t))\mu_2$ , with  $\lambda(t)$  being the mixing factor that controls the proportion of a large step-size  $\mu_1$  and a small step-size  $\mu_2$ .

In order to constrain  $\lambda(t)$  within the interval  $[0, 1]$ , it is computed indirectly by the sigmoid function as

$$\lambda(t) = \text{sigm}[\alpha(t)] = \frac{1}{1 + \exp(-\alpha(t))}, \quad (12)$$

where  $\alpha(t)$  is an auxiliary variable used to control the mixing factor  $\lambda(t)$  in an unconstrained domain, so that  $\lambda(t)$  is always restricted to  $[0, 1]$  through the sigmoid mapping.

To reduce the disturbance of impulsive noises on the adaptation of  $\alpha(t)$ , we define the following robust cost function:

$$J_\alpha(\mathbf{e}(t)) \triangleq \kappa_\sigma(0) - \kappa_\sigma(\|\mathbf{e}(t)\|_2). \quad (13)$$

To update  $\alpha(t)$ , we employ the gradient normalization principle, i.e.,

$$\alpha(t) = \alpha(t-1) - \eta_\alpha \frac{g_\alpha}{|g_\alpha|}, \quad (14)$$

where  $\eta_\alpha > 0$  denotes the learning rate for adjusting  $\alpha(t)$ , and  $g_\alpha \triangleq \frac{\partial J_\alpha(\mathbf{e}(t))}{\partial \alpha(t-1)}$  is the corresponding gradient whose explicit expression is given by

$$g_\alpha = -\frac{(\mu_1 - \mu_2)\lambda(t-1)(1 - \lambda(t-1))}{\|\mathbf{y}(t-1)\|_2^2} \kappa_\sigma(\|\mathbf{e}(t)\|_2) \times \kappa_\sigma(\|\mathbf{e}(t-1)\|_2) \Re\{\mathbf{e}^H(t)\mathbf{e}(t-1)\mathbf{y}^H(t-1)\mathbf{y}(t)\}. \quad (15)$$

By substituting (15) into (14), we can update  $\alpha(t)$  by

$$\alpha(t) = \alpha(t-1) + \eta_\alpha \text{sgn}\left(\Re\{\mathbf{e}^H(t)\mathbf{e}(t-1)\mathbf{y}^H(t-1)\mathbf{y}(t)\}\right). \quad (16)$$

To further suppress abrupt fluctuations,  $\lambda(t)$  is exponentially smoothed as

$$\bar{\lambda}(t) = \xi \bar{\lambda}(t-1) + (1 - \xi)\lambda(t), \quad (17)$$

where  $\bar{\lambda}(t)$  denotes the smoothed value and  $\xi \in (0, 1)$  is the smoothing factor.

Additionally, the signal subspace may change abruptly thus it causes estimation mismatch and degraded tracking performance. To surmount this point, we define the projection energy ratio

$$\rho(t) = \frac{\|\mathbf{W}(t-1)\mathbf{y}(t)\|_2^2}{\|\mathbf{x}(t)\|_2^2}, \quad (18)$$

TABLE 1  
COMPLEXITY COMPARISON OF VARIOUS ALGORITHMS.

| Algorithms              | complexity       |
|-------------------------|------------------|
| PAST [4]                | $O(Nr) + O(r^2)$ |
| MCC-PAST [11]           | $O(Nr) + O(r^2)$ |
| RSTAPS [12]             | $O(Nr^2)$        |
| GDST [4]                | $O(Nr)$          |
| MCC-GDST (proposed)     | $O(Nr)$          |
| CSS-MCC-GDST (proposed) | $O(Nr)$          |

which measures the fraction of the energy of  $\mathbf{x}(t)$  captured by the estimated subspace. To avoid false detection, the running maximum of  $\rho(t)$  is maintained as a reference

$$\rho_{\text{ref}}(t) = \max\{\rho_{\text{ref}}(t-1), \rho(t)\}. \quad (19)$$

After the initial transient, an abrupt change is declared when  $\rho(t) < \tau \rho_{\text{ref}}(t)$ , where  $\tau \in (0, 1)$  is a fixed threshold. In this case,  $\bar{\lambda}(t-1)$  is reset to one, driving  $\bar{\lambda}(t)$  rapidly toward one and enlarging the step-size for fast re-convergence. Meanwhile, we update  $\rho_{\text{ref}}(t) \leftarrow \rho(t)$  to avoid repeated triggering during re-convergence.

**Remark 3:** Table 1 summarizes the complexity of various subspace tracking algorithms, obtained by counting multiplications of the dominant matrix-vector for each iteration or snapshot. As shown in Table 1, compared with the PAST-type and the RSTAPS algorithms, the proposed MCC-GDST and CSS-MCC-GDST algorithms retain inherent low computational cost from the GDST family, and this advantage is further amplified as the number  $r$  of signal sources increases.

## IV. SIMULATION RESULTS

In this section, the proposed CSS-MCC-GDST algorithm is addressed and compared with the existing PAST [4], RSTAPS [12], and MCC-PAST [11] algorithms in both scenarios of subspace estimation and DOA tracking. The additive noise  $n(t)$  is modeled as a SaaS impulsive process with the characteristic exponent  $\alpha = 1.5$  and the scale parameter  $\gamma = 0.1$ .

### A. Subspace Estimation

For evaluation of algorithms, we employ the subspace estimation performance (SEP) [15] [16], which is defined as

$$\text{SEP}(t) = 10 \log_{10} \frac{\text{tr}(\mathbf{W}^H(t)(\mathbf{I}_N - \mathbf{W}_o \mathbf{W}_o^H) \mathbf{W}(t))}{\text{tr}(\mathbf{W}^H(t)(\mathbf{W}_o \mathbf{W}_o^H) \mathbf{W}(t))}, \quad (20)$$

where  $\mathbf{W}(t)$  denotes the estimated subspace. In (20),  $\mathbf{W}_o$  is the reference signal-subspace basis obtained by orthogonalizing the steering matrix  $\mathbf{A}$  in (1), whose entries are independent and identically distributed complex Gaussian random variables with zero-mean and unit-variance. To assess the tracking capability,  $\mathbf{W}_o$  is switched to another independently generated subspace using the same procedure at the midpoint of the iteration. All simulation results are averaged over 100 independent trials.

Fig. 1 compares the considered algorithms under SaaS impulsive noises. Relative to the PAST and GDST algorithms,

other algorithms including the proposed MCC-GDST maintain acceptable performance owing mainly to the fact that they take advantage of some robust strategies such as MCC. By designing the CSS scheme, the proposed CSS-MCC-GDST algorithm effectively solves the problem caused by the constant step-size presented in the MCC-GDST algorithm, thereby achieving a faster convergence rate and higher steady-state accuracy simultaneously. Although the MCC-PAST algorithm can achieve higher steady-state accuracy, it is sensitive to the FF and it may be numerically unstable (e.g., when  $\beta=0.998$ ), while the proposed CSS-MCC-GDST algorithm avoids these flaws. In addition, as one of the GDST family, the CSS-MCC-GDST algorithm has lower per-iteration complexity than the PAST-type algorithms, see statements in Remark 3.

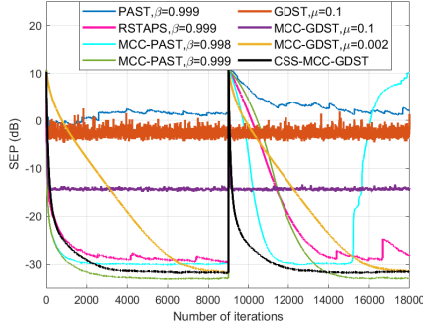


Fig. 1. Comparison of the proposed MCC-GDST and CSS-MCC-GDST algorithms with several competitors under SoS impulsive noises. The selection of some parameters for the algorithms is as follows:  $\epsilon = 2 \times 10^{-3}$ ,  $K = 10$ ,  $\lambda_n = 1$  (RSTAPS);  $\sigma = 2$  (MCC-PAST);  $\sigma = 2$ ,  $\eta_\alpha = 1$ ,  $\alpha(0) = 5$ ,  $\bar{\lambda}(0) = 0.86$  (CSS-MCC-GDST).

### B. DOA Tracking

To assess the practical utility of the algorithm, this subsection addresses the DOA application based on a 20-sensor uniform linear array (ULA). Two narrowband plane-wave sources impinge on the ULA, where the wavelength is normalized to unity and the inter-sensor spacing is set to  $d = 0.5$ . The DOAs are extracted from the estimated subspace using the ESPRIT algorithm. The first DOA is fixed at  $\theta_1 = 10^\circ$ , whereas the second DOA varies with the snapshot index  $t$  as

$$\theta_2(t) = \begin{cases} 40 - 1 \times 10^{-2}t, & 0 \leq t < 400, \\ 36 - 2.5 \times 10^{-1}(t - 400), & 400 \leq t \leq 440, \\ 26 - 7.5 \times 10^{-3}(t - 440), & 440 < t \leq 1000. \end{cases} \quad (21)$$

We run 500 Monte Carlo trials and compute the root mean square error (RMSE) of the time-varying angle  $\theta_2(t)$ , defined as

$$\text{RMSE}(t) = 10 \log_{10} \left( \sqrt{\frac{1}{500} \sum_{m=1}^{500} (\hat{\theta}^{(m)}(t) - \theta_2(t))^2} \right), \quad (22)$$

where  $\hat{\theta}^{(m)}(t)$  denotes the estimate of  $\theta_2(t)$  at the  $t$ -th snapshot in the  $m$ -th trial.

Fig. 2 compares the DOA tracking trajectories of the considered algorithms. As can be seen, the PAST and RSTAPS

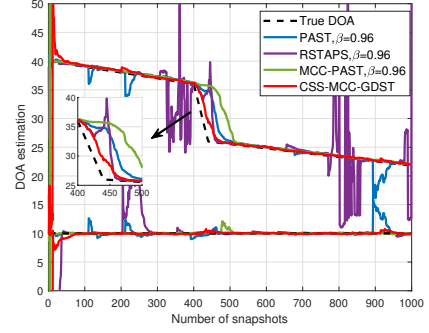


Fig. 2. DOA estimation of the algorithms. The selection of some parameters for the algorithms is as follows:  $\epsilon = 2 \times 10^{-3}$ ,  $K = 10$ ,  $\lambda_n = 1$  (RSTAPS);  $\sigma = 10$  (MCC-PAST);  $\sigma = 10$ ,  $\eta_\alpha = 1$ ,  $\alpha(0) = 5$ ,  $\bar{\lambda}(0) = 0.86$  (CSS-MCC-GDST).

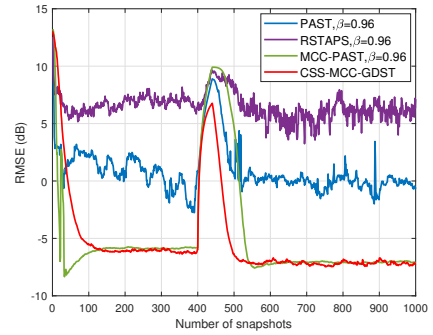


Fig. 3. RMSE comparison for the algorithms.

algorithms exhibit pronounced fluctuations, indicating sensitivity to impulsive disturbances. The MCC-PAST algorithm yields smooth estimates, but it suffers a noticeable lag during rapid DOA variations due to using the fixed FF. By contrast, the proposed CSS-MCC-GDST algorithm tracks the true DOA more closely with reduced fluctuations and shorter transients.

To quantitatively evaluate the performance of the algorithms in Fig. 2 for DOA, Fig. 3 depicts the RMSE evolution for  $\theta_2(t)$ . It can be seen that the proposed CSS-MCC-GDST algorithm not only reaches the same steady-state RMSE (i.e., accuracy) as the MCC-PAST algorithm but also has better tracking behavior for the case of fast change of DOA.

### V. CONCLUSION

In this work, to overcome the existing shortcomings of the GDST algorithm, we addressed two improvements. By incorporating the MCC into GDST, the MCC-GDST algorithm was proposed to achieve good robustness against impulsive noises. Then, to obtain fast convergence and high steady-state accuracy, the CSS-MCC-GDST algorithm was further developed by designing a CSS scheme instead of the existing constant step-size. Finally, simulations in terms of subspace estimation and DOA estimation were conducted to demonstrate that the proposed algorithms outperformed their competitors.

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